

Learning Control

The printer setup

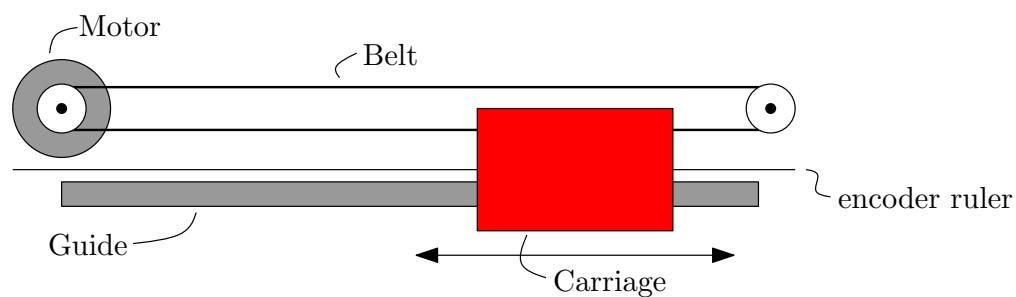
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Modeling the printer setup

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Schematic overview:



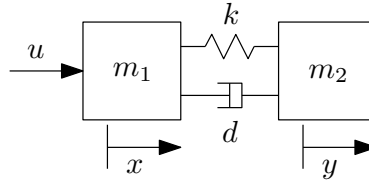
Specifications:

- ▶ Motor is current controlled, input range ± 2.5 V
- ▶ Carriage position is measured by a 600 CPI optical encoder

Modeling the printer setup

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Proposed model:



Variables:

- u , motor torque
- x , motor position (not measured in the physical setup)
- y , carriage position

Equations of motion:

$$m_1 \ddot{x} = k(y - x) + d(\dot{y} - \dot{x}) + u$$

$$m_2 \ddot{y} = k(x - y) + d(\dot{x} - \dot{y})$$

Modeling the printer setup

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Physical parameters:

$$m_1 = 1.0 \cdot 10^{-3} \text{ kg}$$

$$m_2 = 9.0 \cdot 10^{-2} \text{ kg}$$

$$k = 2.35 \cdot 10^2 \text{ Nm}^{-1}$$

$$d = 1.2 \cdot 10^{-1} \text{ kgs}^{-1}$$

The model in state space form:

$$\dot{z} = Az + Bu$$

$$y = Cz + Du$$

State:

$$z = \begin{bmatrix} x & \dot{x} & y & \dot{y} \end{bmatrix}^T$$

With:

$$A = \begin{bmatrix} 0 & 1 & 0 & 0 \\ -m_1^{-1}k & -m_1^{-1}d & m_1^{-1}k & m_1^{-1}d \\ 0 & 0 & 0 & 1 \\ m_2^{-1}k & m_2^{-1}d & -m_2^{-1}k & -m_2^{-1}d \end{bmatrix}$$

$$B = [0 \quad m_1^{-1} \quad 0 \quad 0]^T$$

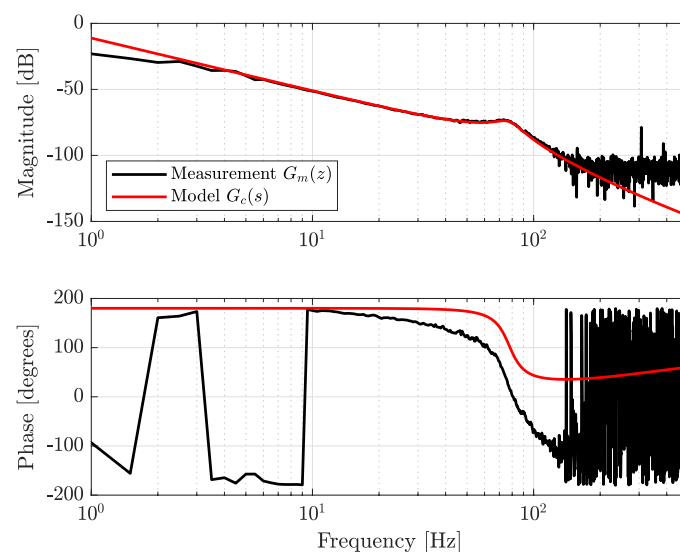
$$C = [0 \quad 0 \quad 1 \quad 0]$$

$$D = 0$$

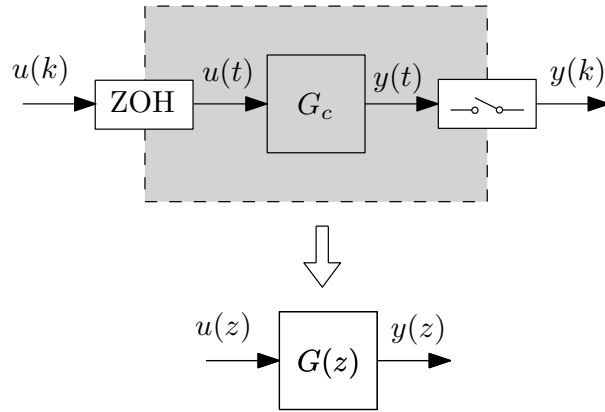
Transfer function $y = G(s)u$:

$$G(s) = \frac{ds + k}{m_1 m_2 s^4 + d(m_1 + m_2)s^3 + k(m_1 + m_2)s^2}$$

Frequency response measurement and continuous time model:

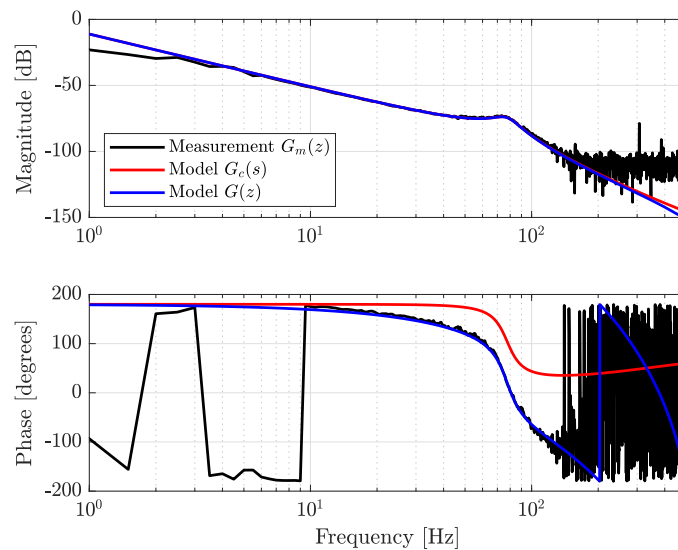


Sampling and holding:



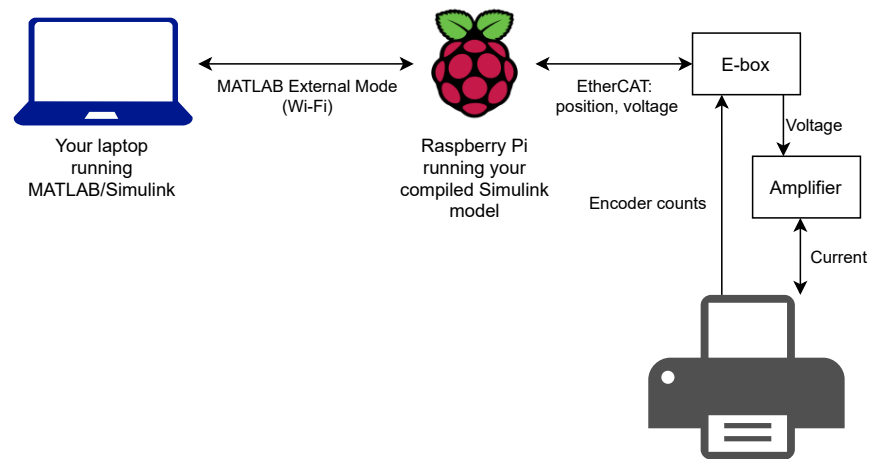
In matlab using "c2d(Gc,Ts,'zoh')", see Franklin et al. (2008).

Frequency response measurement and discrete-time model:



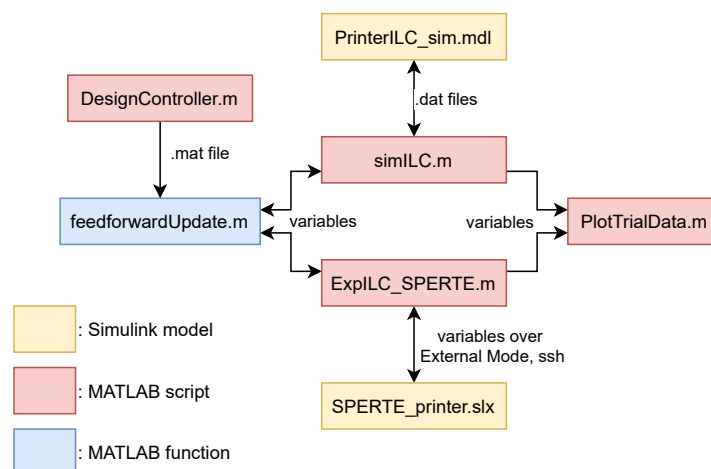
Hardware overview

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Software overview

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Make sure you install the correct Matlab version and required toolboxes before the experiments! See SPERTE Installation PDF on Canvas.

Franklin, G., Powell, J. & Emami-Naeini, A. (2008), *Feedback Control of Dynamic Systems*, Pearson.